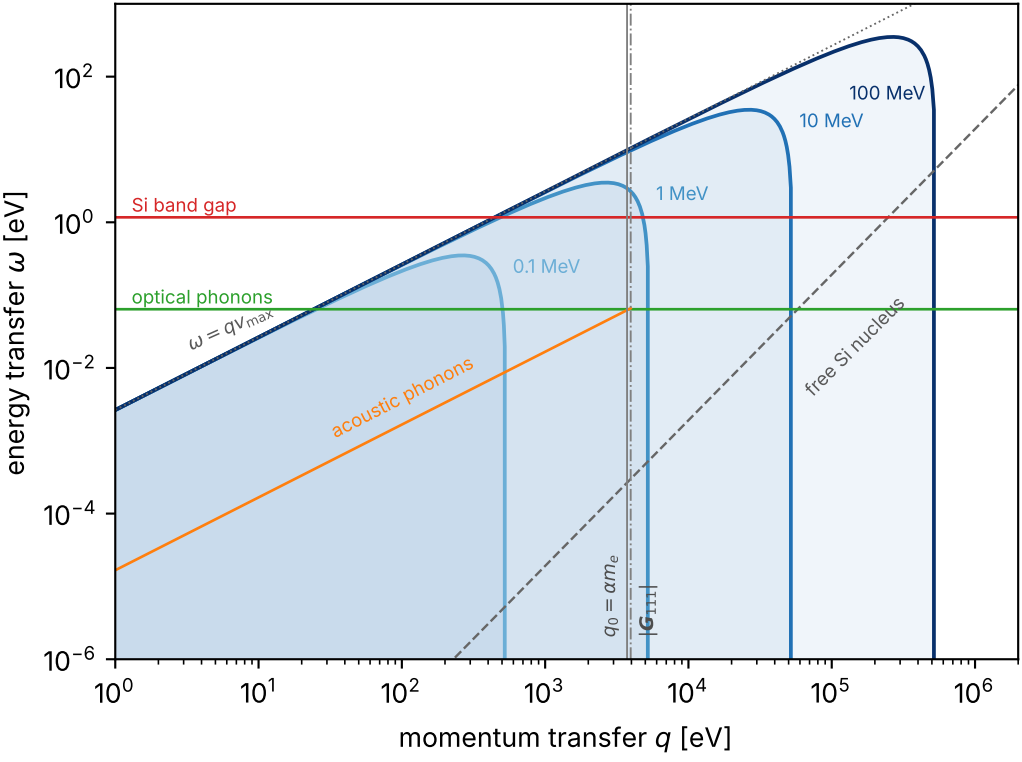

energy transfer ω [eV]
momentum transfer q [eV]
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
100 MeV
10 MeV
1 MeV
0.1 MeV
Si band gap
optical phonons
$\omega = qv_{max}$
acoustic phonons
free Si nucleus
$q_0 = \alpha m_e$
$|\boldsymbol{G}_{111}|$